Data-driven Robust Bipartite Weighted Matching with Stochastic Features

Ziyue Li, Leo Li , Wyame Benslimane
May 10, 2023

1 Introduction

Over the past few decades, the problem of Online Bipartite Matching has garnered significant attention in academic research and found a broad range of practical applications in real-world systems [4]. The primary objective of this problem is to design efficient algorithms that can match sequential arrivals to available resources in real-time without sacrificing future decision-making capabilities or overall revenue generation potential.

The following scenario presents a challenging Online Bipartite Matching problem encountered by a travel agency that sells flight tickets to different customer types, namely business travelers, budget travelers, and families. Each customer type has a unique set of preferences, including preferred airlines, departure times, and layovers, which must be satisfied in real-time while maximizing the revenue generated. The travel agency has access to a large database of flight tickets, each characterized by varying departure and arrival times, airlines, and prices. Specifically, business travelers prefer direct flights with premium airlines, budget travelers opt for low-cost airlines with layovers, and families prefer flights with convenient departure and arrival times and minimal layovers. Upon the arrival of a new customer, the travel agency must match them with an available flight ticket that meets their preferences and budget while maximizing revenue. As more customers arrive, the travel agency must continuously update the matching of flight tickets to customers, while adhering to their preferences and budget constraints.

In practice, the Online Bipartite Matching problem encounters a stochastic nature where the customer type is unknown before their arrival. This feature poses significant challenges to the matching algorithms. To address this issue, an adversarial approach can be taken initially, assuming nothing about the type of arrivals being encountered. Several algorithms have been developed for this scenario, providing strict guarantees on the worst-case scenarios. For instance, the ranking algorithm achieves a competitive ratio of 1-1/e [3]. However, such algorithms can be overly cautious and simplistic, neglecting the prior knowledge that can be utilized to derive a near-optimal solution for instances where patterns can be inferred.

Alternatively, a stochastic approach can be implemented, incorporating prior knowledge of the distribution of arrivals to make informed real-time decisions. This approach aims to capture different patterns and utilize the knowledge to develop an online algorithm that achieves near-optimal solutions. Recent advancements in learning models have enabled the utilization of collected data to enhance predictions by building a demand model and using inferred information to optimize the matching algorithm.

In the context of the travel agency scenario, an approach called Online Bipartite Matching with Stochastic Features can be utilized to optimize the matching process between the travel agency and customers in real-time. The primary objective of this algorithm is to find the best matching while accounting for the stochastic features and delivering an efficient and cost-effective service. However, constructing such stochastic demand models can potentially result in a misrepresentation of real demand, leading to a less robust decision-making algorithm that may produce suboptimal decisions.

Therefore, it is crucial to investigate the impact of data-driven demand prediction on the decision model and design algorithms that can mitigate the adverse effects of these predictions when making online decisions. Our project aims to achieve two objectives. Firstly, we will explore how modeling errors in demand prediction affect the performance of the online matching algorithm. We intend to derive a lower bound on the model to quantify the extent of these errors accurately. Secondly, we will leverage tools in the field of robust optimization [1] to develop an algorithm that can minimize the impact of these errors on the decision-making process. This will involve developing more robust algorithms that can make decisions that are less sensitive to the predictions made by the stochastic demand models. Overall, our goal is to develop more effective and reliable online matching algorithms that can make real-time decisions with greater accuracy and efficiency.

2 Model

2.1 Preliminaries

In the context of estimating parameters from a data-driven perspective, it is essential to assess the accuracy of our estimator based on the total number of historical data. To this end, we can utilize the nonasymptotic version of the Glivenko-Cantelli theorem, which provides a bounding result for the empirical estimator.

Lemma 2.1 (Glivenko-Cantelli Theorem [5])**.** *Assume that $X_1, \cdots, X_n$ are independent and identically distributed random variables in $\mathbb{R}$ with common cumulative distribution function F. The empirical distribution function for $X_1, \cdots, X_n$ is F_n. Then,*

$$\mathbb{P}\left(\|F_n - F\|_\infty \geq \sqrt{\frac{8\log(n+1)}{n}} + \epsilon\right) \leq \exp\left(-\frac{n\epsilon^2}{2}\right) \tag{1}$$

Corollary 2.2. *Assume that $X_1, \cdots, X_n$ are independent and identically distributed random variables in $\mathbb{R}$ with common probability density function P. The empirical density function for $X_1, \cdots, X_n$ is P_n. Then,*

$$\mathbb{P}\left(\|P_n - P\|_\infty \geq \sqrt{\frac{32\log(n+1)}{n}} + \epsilon\right) \leq \exp\left(-\frac{n\epsilon^2}{2}\right) \tag{2}$$

To calculate the probability that resources do not exceed the inventory, we may apply the standard tool of the Chernoff bound.

Lemma 2.3 (Chernoff Bound)**.** *Given independent Bernoulli random variables $X_1, \cdots, X_T$ with mean $x_1, \cdots, x_T$. Let $\sum_{t \le T} x_t \le \mu$. Then*

$$\mathbb{P}\left(\sum_{1 \le t \le T} X_t \ge (1+\delta)\mu\right) \le e^{-\frac{\delta^2}{2+\delta}\mu} \tag{3}$$

2.2 Resource Allocation Framework

We consider the following framework of online matching problems, which involves a set of I resources. Each resource $i \in I$ has a fixed capacity c_i, and there are T requests belonging to a unique type $z \in Z$. These requests can be satisfied by assigning them to a specific resource i, denoted by x_{iz}, which generates a corresponding revenue amount w_{iz}. Let n_z be the expected number of arrivals with type z, then it can be shown that the offline LP optimization problem can be formulized as Problem 4.

$$\begin{aligned} \max \quad & \sum_i \sum_z w_{iz} n_z x_{iz} \\ s.t. \quad & \sum_{z \in Z} n_z x_{iz} \le c_i, \quad \forall i \in I \\ & \sum_{i \in I} x_{iz} \le 1, \quad \forall z \in Z \\ & 0 \le x_{iz} \le 1 \end{aligned} \tag{4}$$

For our model, we assume that the total number of arrivals T, the arrival type z with its corresponding revenue w_{iz}, and the resource capacity c_i are known. Also, we assume that the distribution of types remains unchanged. If we have collected n historical data, we can then estimate P_z using the empirical density $\hat{P}_z$. Directly applying 2 yields the following concentration result.

Theorem 2.4. *If we use n historical data to estimate the probability density, then the estimation error satisfies that*

$$\mathbb{P}\left(\|\hat{P}_z(n) - P_z\|_\infty \ge \sqrt{\frac{32\log(n+1)}{n}} + \epsilon\right) \le \exp\left(-\frac{n\epsilon^2}{2}\right) \tag{5}$$

Or equivalently, $P_z \in \hat{P}_z(n) \pm (\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$ with probability at least $1 - \exp(-\frac{n\epsilon^2}{2})$.

2.3 Robust Optimization Model

Let the uncertainty set to be $\mathcal{B}(\hat{n}_z, \rho_0) = [\hat{n}_z - \rho_0, \hat{n}_z + \rho_0]$. We solve the robust optimization (RO) problem 6.

$$
\begin{aligned}
\max \min_{n_z \in \mathcal{B}(\hat{n}_z, \rho_0)} \quad & \sum_i \sum_z w_{iz} n_z x_{iz} \\
s.t. \quad & \sum_{z \in Z} n_z x_{iz} \leq c_i, \quad \forall i \in I, \quad \forall n_z \in \mathcal{B}(\hat{n}_z, \rho_0) \\
& \sum_{i \in I} x_{iz} \leq 1, \quad \forall z \in Z \\
& 0 \leq x_{iz} \leq 1
\end{aligned} \tag{6}
$$

It can be shown that problem 6 is equivalent to problem 7 below.

$$
\begin{aligned}
\max \quad & \sum_i \sum_z w_{iz} (\hat{n}_z - \rho_0) x_{iz} \\
s.t. \quad & \sum_{z \in Z} (\hat{n}_z + \rho_0) x_{iz} \leq c_i, \quad \forall i \in I \\
& \sum_{i \in I} x_{iz} \leq 1, \quad \forall z \in Z \\
& 0 \leq x_{iz} \leq 1
\end{aligned} \tag{7}
$$

We assume the optimal solution of problem 7 is $\hat{x}_{iz}$, and the optimal solution of the offline problem 4 is x^*_{iz}. To derive the optimal competitive ratio, we pick the radius ρ_0 of the uncertainty set $\mathcal{B}(\hat{n}_z, \rho_0)$ to be $\rho_0 = T(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$, as suggested by theorem 2.4. Here ϵ is specified by the user. More on this in the following sections.

3 Competitive Ratio

3.1 Probability of Out-of-stock

Let $X_{it} = I(i \to t) \in \{0, 1\}$ to be the indicator of whether the resource i is matched to arrival t in our algorithm, then the expected matching number to resource i is

$$
\text{Mean}(\sum_t X_{it}) \leq \sum_z n_z \hat{x}_{iz} = \sum_z (n_z + \rho_0) \hat{x}_{iz} \frac{n_z}{n_z + \rho_0} \leq c_i \max_z \frac{n_z}{n_z + \rho_0} = c_i \rho \tag{8}
$$

where $\rho = \max_z \frac{n_z}{n_z + \rho_0}, \rho_0 = T(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$. If we let $p_{\max} = \max_z P_z$, then

$$
\rho = \frac{1}{1 + \frac{\rho_0}{T p_{\max}}} = \frac{1}{1 + \frac{\sqrt{\frac{32\log(n+1)}{n}} + \epsilon}{p_{\max}}}, \qquad p_{\max} \in (0, 1) \tag{9}
$$

Therefore,

$$
\text{Mean}(\sum_t X_{it}) \leq c_i \rho = \frac{c_i}{1 + \frac{\sqrt{\frac{32\log(n+1)}{n}} + \epsilon}{p_{\max}}} \tag{10}
$$

Noticed that the mean value of resource i consumed is already less than c_i. This is the direct result of robustness. This means that robust optimization is equivalent to attenuation in this example.

Proposition 3.1. *RO is equivalent to attenuation.*

We pick $\delta_i = \frac{1}{\rho} - 1 = \frac{\sqrt{\frac{32\log(n+1)}{n}}+\epsilon}{p_{\max}}$. By lemma 2.3, for resource i, the out-of-inventory probability is calculated below.

$$\begin{aligned}
\mathbb{P}\left(\sum_t I(i \to t) \geq (1+\delta_i)c_i\rho\right) &= \mathbb{P}\left(\sum_t I(i \to t) \geq c_i\right) \leq \exp\left(-\frac{\delta_i^2}{2+\delta_i}c_i\rho\right) \\
&\leq \exp\left(-\frac{(\frac{1}{\rho}-1)^2}{1+\frac{1}{\rho}}c_i\rho\right) = \exp\left(-\frac{(\rho-1)^2}{\rho+1}c_i\right) \\
&= \exp\left(-c_i \cdot (\rho + 1 + \frac{4}{\rho+1} - 4)\right) \\
&= \exp\left(-c_i \cdot \left(2 - O(\frac{\sqrt{\frac{\log(n+1)}{n}}+\epsilon}{p_{\max}}) + \frac{4}{2 - O(\frac{\sqrt{\frac{\log(n+1)}{n}}+\epsilon}{p_{\max}})} - 4\right)\right) \\
&= \exp\left(-c_i \cdot O(\frac{\sqrt{\frac{\log(n+1)}{n}}+\epsilon}{p_{\max}})\right)
\end{aligned} \tag{11}$$

where we used the fact that $\rho = 1 - O(\frac{\sqrt{\frac{\log(n+1)}{n}}+\epsilon}{\rho})$ when $n \to \infty$ and $\epsilon \to 0$.

3.2 Competitive Ratio Given No Out-of-stock

Consider the optimal offline objective value OPT. Let x_{itz} be the probability that t is matched to i in OPT conditional on type z arrival at t, then it can be shown that $\tilde{x}_{iz} = \frac{1}{n_z}\sum_t F_t(z)x_{itz}$ is a feasible solution to problem 4 with objective equal to E[OPT]. Therefore, if the LP solution of the offline problem 4 is x^*_{iz}., $\sum_{i,z} w_{iz}n_z x^*_{iz}$ gives an upper bound of E[OPT].

$$\sum_{i,z} w_{iz}n_z x^*_{iz} \geq \text{E[OPT]} \tag{12}$$

Noticed that $\sum_z n_z \cdot x^*_{iz} \leq c_i$, we have $\sum_z(\hat{n}_z + \rho_0)x^*_{iz}\frac{n_z}{\hat{n}_z+\rho_0} \leq c_i$. Therefore, $x^*_{iz} \cdot \frac{n_z}{\hat{n}_z+\rho_0}$ is a feasible solution of problem 7.

The expected object of the robust optimization problem is

$$\begin{aligned}
\text{E[RO]} &\geq \sum_{i,z} w_{iz}(\hat{n}_z - \rho_0)x^*_{iz}\frac{n_z}{\hat{n}_z+\rho_0} \geq \min\frac{\hat{n}_z-\rho_0}{\hat{n}_z+\rho_0}\text{E[OPT]} \\
&\geq \min\frac{n_z - 2\rho_0}{n_z}\text{E[OPT]} = \min(1 - 2\frac{\rho_0}{n_z})\text{E[OPT]}
\end{aligned} \tag{13}$$

If we let $p_{\min} = \min p_Z$, then

$$\text{E[RO]} \geq \left(1 - 2\frac{T(\sqrt{\frac{32\log(n+1)}{n}}+\epsilon)}{p_{\min}T}\right)\text{E[OPT]} = \left(1 - O(\frac{\sqrt{\frac{\log(n+1)}{n}}+\epsilon}{p_{\min}})\right)\text{E[OPT]} \tag{14}$$

3.3 Main Result

Theorem 3.2. *If we use the solution x^*_{iz} of problem 7 as the allocation rule, i.e., for arrival t, if it's type z, then independently sample a resource $i_t \sim (x^*_{iz})$. Match t to i_t if available, otherwise ignore t. Then with probability*

$$\prod_i \left(1 - \exp\left(-c_i \cdot O(\frac{\sqrt{\frac{\log(n+1)}{n}} + \epsilon}{p_{\max}})\right)\right) \tag{15}$$

the algorithm achieves a competitive ratio

$$CR \geq 1 - O\left(\frac{\sqrt{\frac{\log(n+1)}{n}} + \epsilon}{p_{\min}}\right) \tag{16}$$

3.4 Discussion

3.4.1 Implicit Assumptions

To make sure that $n_z - 2\rho_0 \geq 0$ in equation (13), we have the following assumption.

Assumption 3.3.

$$p_{\min} \geq 2(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon) \tag{17}$$

3.4.2 Parameter ϵ and Competitive Ratio

It can be observed from theorem 3.2 that the probability guarantee increases as the value of ϵ increases, whereas the competitive ratio decreases. This is due to the fact that a larger radius of the uncertainty set results in a higher probability of the true value of p_z lying within the uncertainty set, and a lower likelihood of exceeding the inventory. However, it should be noted that achieving greater robustness comes at the expense of reduced profit levels. Hence, the choice of parameter ϵ should be based on the user's preference. If a high probability guarantee for achieving the competitive ratio is desired, a larger value of ϵ may be selected. On the other hand, if a better competitive ratio with a reasonable probability guarantee is required, a smaller value of ϵ may be preferred.

3.4.3 Distribition of Types and $p_{\max}, p_{\min}$

The performance of the main result highly depends on the distribution of types, especially $p_{\max}$ and $p_{\min}$. In assumption 3.3 we require that $p_{\min}$ is not too small. Substitute (17) into theorem 3.2 yields the result that the competitive ratio is at least $\frac{1}{2}$. If we have more data, we can have near-optimal CR. The assumption below gives the desired result.

Assumption 3.4. *When $n = 1/O(p_{\min}^{2+\gamma})$, where $\gamma > 0$, and $\epsilon = o(p_{\min})$, we may achieve a near-optimal competitive ratio.*

The discussion of $p_{\max}$ and probability guarantee is more complicated, because of the appearance of c_i and the exponential form. By the pigeonhole's principle, we have $1/Z \leq p_{\max} \leq 1$. Assuming that Z is finite and not too large, which means that $p_{\max}$ is not so small, we have the following result.

Corollary 3.5. *The probability p_{AI} of adequate inventory for any resource i satisfies that*

$$p_{AI} = \prod_i \left(1 - \exp\left(-c_i \cdot O(\sqrt{\frac{\log(n+1)}{n}} + \epsilon)\right)\right) \tag{18}$$

Assumption 3.6. *Assume that $\frac{1}{c_i} = O\left((\sqrt{\frac{\log(n+1)}{n}} + \epsilon)^{1+\alpha}\right)$. When $\alpha > 0$, the probability p_{AI} tends to 1.*

Theorem 3.7. *Assume assumption 3.3, 3.4 and 3.6 hold. i.e.,*

- $p_{\min} \geq 2(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$,
- $n = 1/O(p_{\min}^{2+\gamma})$, *where* $\gamma > 0$,
- $\epsilon = o(p_{\min})$,
- $\frac{1}{c_i} = O\left((\sqrt{\frac{\log(n+1)}{n}} + \epsilon)^{1+\alpha}\right)$, *where* $\alpha > 0$,

we get the near-optimal competitive ratio.

4 Comparison with Other Methods

In addition, there exist alternative online algorithms that offer near-optimal solutions for resource allocation problems involving an unknown distribution. [2] introduce a set of resilient algorithms for resource allocation problems that entail independently drawn requests from an unknown distribution in every time step. Their proposed approach entails estimating the upper bound of the offline Linear Programming (LP) and assigning resources based on the resulting value, rather than learning the empirical probability as in our method. Their algorithm attains a competitive ratio (CR) of $1 - O(1/\epsilon)$ with a probability of $1 - \epsilon$. This achievement relies on a set of assumptions.

- $\epsilon \geq \frac{1}{n}$,
- $\gamma = O(\frac{\epsilon^2}{\log(I/\epsilon)})$, where $\gamma = \max\{\{\frac{1}{c_i}\}, \{\frac{w_{iz}}{\mathrm{OPT}}\}\}$.

Compared with their algorithm, our method has the following features:

- Our model and analysis give a non-asymptotic result of the relationships of CR, probability guarantee, n, ϵ, T, and c_i. The results are more detailed and elegant.
- We do not require $\epsilon \geq \frac{1}{n}$. ϵ can be quite small. And it can be specified by users. As ϵ goes smaller, CR goes larger but with a smaller probability guarantee.
- The type distribution is of significant consequence, particularly with regard to $p_{\min}$. Our analysis necessitates that $p_{\min} \geq 2(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$. This requirement seems to contrast with that of [2], which instead places restrictions on the ratio of w_{iz} to the optimal solution, i.e.,

w_{iz}/OPT is small enough. Although $p_{\min}$ may be very small, the corresponding type may generate a negligible amount of revenue, rendering it dispensable. Thus, in some sense, both approaches are consistent with one another.

As inspired by [2], we may propose a new algorithm that performs better when T is at a higher order of n, or even when n is small. The algorithm 1 is shown below. This generalization aids in the development of the algorithm (1) in cases where historical data may be insufficient, but a substantial number of arrivals is anticipated. In such scenarios, it is feasible to exclude the initial portion of arrivals in order to obtain parameter estimates, followed by serving customers at varying stages and iteratively updating the parameters subsequent to each stage.

Algorithm 1 Periodic Calibration Model

Input Capacities c_i, total number of requests T ($T \gg n$)
Output The online allocation algorithm

```
Set number of stages l (maybe correlated with error ε)
Stage 0: learn parameter n_z(0) based on historical data n and T/l arrival, but discard all requests.
for Stage j from 1 to l − 1 do
    Customers: T/l, set resources: c_i/(l − 1)
    Apply Robust LP to allocate resources
    Update: n_z(j) ← Empirical Estimator
```

5 Experiment

In this section, we will discuss the design of our experiments, the datasets we used, and the results we obtained. Our experiment aims to evaluate the effectiveness of the proposed optimization method for solving the problem of bipartite weighted matching with stochastic arrival using the previously proposed methods. We wanted to examine how different resource capacities, arrival types and training datapoints affect the system's overall efficiency and performance. To conduct the experiment, we generated a set of synthetic data representing different scenarios of resource allocation then applied the proposed optimization method to the generated data sets.

5.1 Description of our simulator:

To conduct the experiment, we designed a synthetic data generation simulator. The simulator can generate random instances of an online bipartite matching problem with stochastic arrivals. The simulator can generate resources, arrival types, and arrival sequences that follow specific distributions and weights.

In our experiment, we used three main classes: Resources, Arrival Type, and Arrivals. The Resources class represents the resources available for the simulation and includes attributes such as capacity and the number of resources. The Arrival Type class represents the distribution of arrival types in the simulation and includes attributes such as the number of arrival types and the distribution and weights of the edges of the graph. Finally, the Arrivals class allows us to generate different arrival sequences that follow the distribution defined by the Arrival Type class.

(a) We have 10 resources(blue) and 5 arrival types (blue).

(b) We have 5 resources(blue) and 10 arrival types (blue).

Figure 1: Different generated instances, the intensity of the edges represents the weight of each edge.

Using the instance created, we are able to implement the algorithm described above. We first start by generating our training data that is used to estimate the probability distribution of our arrivals type. We use a sample average approximation to estimate the probability distribution of our algorithm.This involves generating a large number of arrival sequences and computing the average frequency of each arrival type. Using the obtained estimate, we focused on the performance of two specific allocation algorithms: the independent rounding algorithm and the attenuated rounding algorithm. After running both algorithms, we compare the results of our algorithms with the optimal allocation that can be obtained by solving the linear programming optimization problem. This allows us to evaluate the performance of our algorithms and determine if they are effective for solving the problem of bipartite weighted matching with stochastic arrivals.

5.2 Description of our experiment:

In order to assess the performance of our proposed method, we run different experiments by varying our parameters to evaluate how each parameter would affect the results of our algorithm We run 4 different experiments:

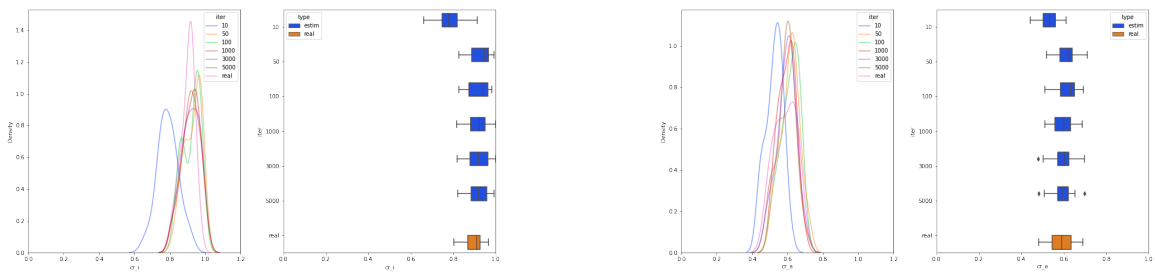
- Changing the instance: we want to evaluate the performance of our approach on both simple and complicated instances.
- Changing the training size: this experiment allows us to determine the impact of the size of our training dataset on the results of the matching algorithm
- Changing the length of the arrival sequences: The purpose is to evaluate how does increasing the length of the arrival sequences while keeping the number of resources fixed to determine the algorithm's efficiency in handling larger datasets
- Changing the capacity of resources: We want to evaluate the effect of the resource capacity on our approach.

These experiments allowed us to gain insights into the strengths and limitations of our approach and determine the optimal parameters for achieving the best results.

5.3 Results for a simple instance:

We first start by generating a simple instance that contains 10 resources and 5 arrival types. For the first experiment, we change the number of training data points:

5.3.1 Experiment 1: The effect of varying the number of training points on a simple instance



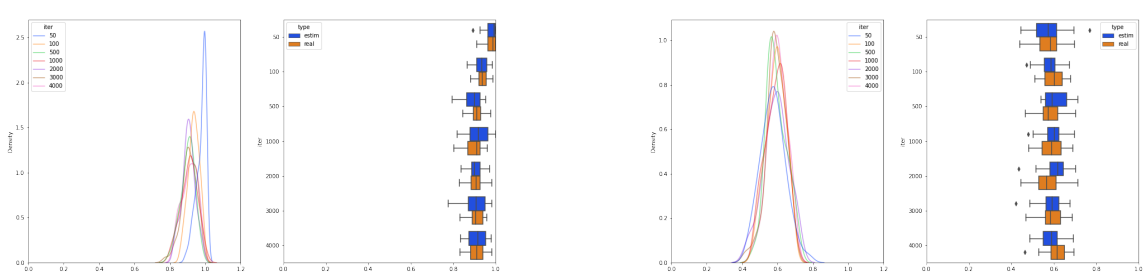
(a) Independent rounding algorithm results when varying the number of training data points

(b) Attenuated rounding algorithm results when varying the number of training data points

Figure 2: Experiment 1

As we can see from the figure (Figure 2) that the performance on the simple instance improves when using a larger dataset, likely as a result of increased prediction accuracy of our distribution. We also see that for the simple instance, the independent rounding algorithm results in near-optimal performance while we can't make the same observation when using the attenuated rounding algorithm.

5.3.2 Experiment 2: The effect of varying the length of the arrival sequence on a simple instance



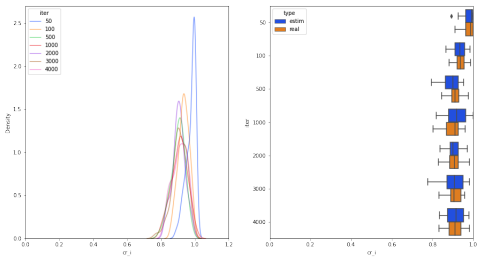
(a) Independent rounding algorithm results when varying the number of arrivals

(b) Attenuated rounding algorithm results when varying the number of arrivals

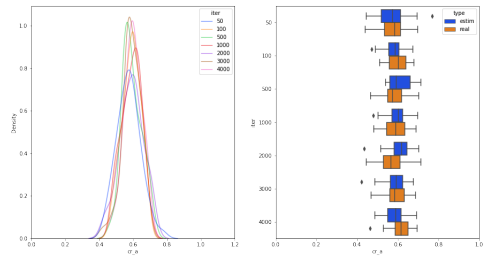
Figure 3: Experiment 2

We can see that for smaller arrival sequences and fixed capacity, the performance of the independent rounding algorithm is comparable to the optimal allocation that we can get if we solve the problem offline. However, we can see that for smaller arrival sequences we get better performances.

5.3.3 Experiment 3: The effect of varying the resources' capacity on a simple instance



(a) Independent rounding algorithm results when varying resources' capacity



(b) Attenuated rounding algorithm results when varying resources' capacity

The experiment results complement the theory as the larger the resource capacity the better the performance of our algorithm.

5.4 Results for a more complicated instance:

We complicate our instance by having the arrival types be very picky regarding the resources that they can be matched with and by having more arrival types and we repeat the same experiment.

5.4.1 Picky arrivals:

The generated instance looks like this:

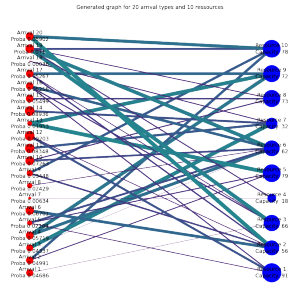
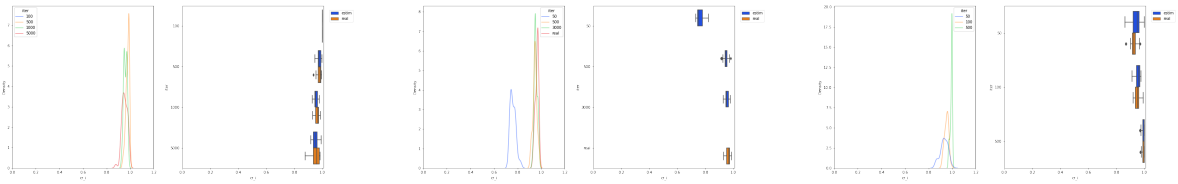


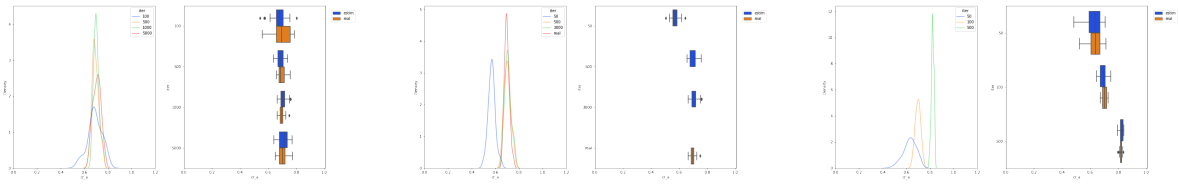
Figure 5: Generated instances

We repeat the same 3 experiments on the previous generated instance and get the following results:



(a) Varying training data-points (b) Varying arrival sequence length (c) Varying capacity of resources

Figure 6: Independent rounding algorithm results



(a) Varying training data-points (b) Varying arrival sequence length (c) Varying capacity of resources

Figure 7: Attenuated rounding algorithm results

We can see that our algorithm exhibits the same properties as the simple instance, which makes our methods more robust.

6 Conclusion

This report presents a novel approach for solving the online bipartite matching problem under conditions where the distribution of arrival types is unknown. To address this challenge, we propose a robust optimization method that can achieve near-optimal solutions by considering worst-case scenarios.

To evaluate the efficacy of our proposed approach, we conducted a series of numerical experiments, comparing it to traditional methods and alternative algorithms. The outcomes of both theoretical derivations and numerical experiments demonstrate that the proposed robustness significantly enhances the performance of the algorithm when parameters are unknown.

Moreover, we compare our approach to other algorithms and generalize our model to cases when sufficient historical data is not available but we anticipate a large number of arrivals. In addition, we investigate the impact of the distribution of arrival types, particularly focusing on the minimum probability $p_{\min}$. Our analysis necessitates that $p_{\min} \geq 2(\sqrt{\frac{32\log(n+1)}{n}} + \epsilon)$, where n is the total number of customers, and ϵ is the parameter that determines the level of uncertainty in the problem.

In conclusion, our study provides valuable insights into data-driven online matching by demonstrating the effectiveness and robustness of our proposed approach in addressing the online bipartite matching problem with unknown arrival type distributions.

References

[1] Aharon Ben-Tal, Laurent El Ghaoui, and Arkadi Nemirovski. *Robust optimization*, volume 28. Princeton university press, 2009.

[2] Nikhil R Devanur, Kamal Jain, Balasubramanian Sivan, and Christopher A Wilkens. Near optimal online algorithms and fast approximation algorithms for resource allocation problems. *Journal of the ACM (JACM)*, 66(1):1–41, 2019.

[3] Richard M Karp, Umesh V Vazirani, and Vijay V Vazirani. An optimal algorithm for on-line bipartite matching. In *Proceedings of the twenty-second annual ACM symposium on Theory of computing*, pages 352–358, 1990.

[4] Aranyak Mehta et al. Online matching and ad allocation. *Foundations and Trends® in Theoretical Computer Science*, 8(4):265–368, 2013.

[5] Martin J Wainwright. *High-dimensional statistics: A non-asymptotic viewpoint*, volume 48. Cambridge university press, 2019.